

WELCOME

GET STARTED

INTRODUCING

Label-Blend Studio, An Automated augmentation & labelling Suite

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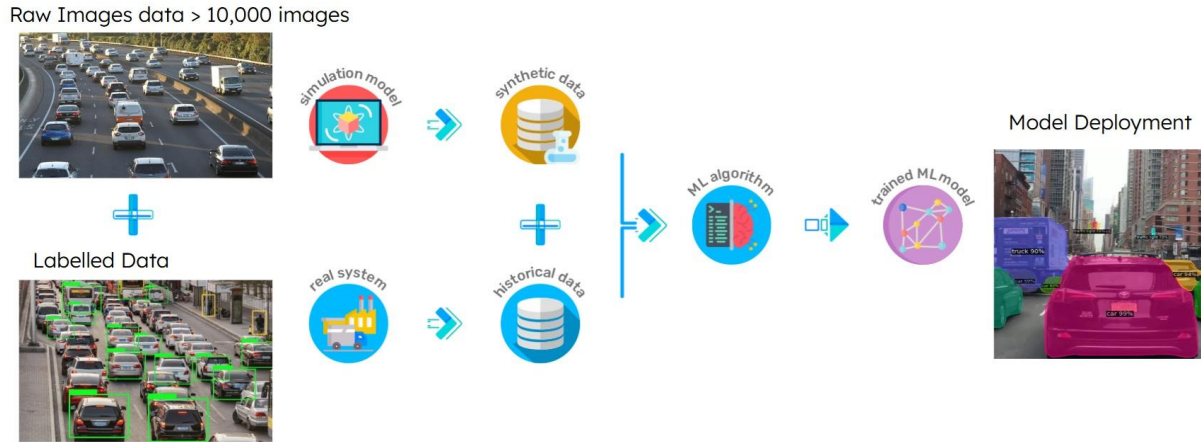
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Major Project / Year 2K25, 7th Sem

Problem Statement

As we know , in supervised learning, training AI models requires large, high-quality labeled datasets, but collecting and labelling real-world images manually is time-consuming, costly, and often sometimes impractical. This causes data scarcity and underfitting. Therefore, there's need of an automated system that can labelled the data for us to reduce model biasing and overfitting problems also it should create diverse datasets so the model give as accurate and useful results.

Computer vision AI model Training approach for supervised machine learning:-



To solve the above problem, we can use , synthetic dataset as reference to real world, Let's understand ..

Large-scale synthetic data generation involves creating a huge numbers of data like images, sound text, statistical data etc using labelling and designing softwares or by using another AI models. but in this project this synthetic data generation means of creating the huge amount of **realistic, computer-generated images** using to train AI models.

Solution

Introducing - Label-Blend Studio, our major project that provides an automated desktop-based system that generates large no. of synthetic images by merging cutouts with dynamic backgrounds and applies random augmentation and lighting.

It automatically creates YOLO-compatible bounding box or segmentation labels, enabling fast, scalable generation of high-quality datasets for AI model training.

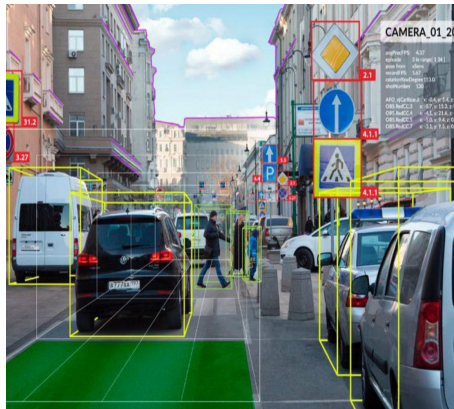
Here are some common problems arrived in traditional image data preparation.

PROBLEMS

1. Data scarcity in image segmentation labelling

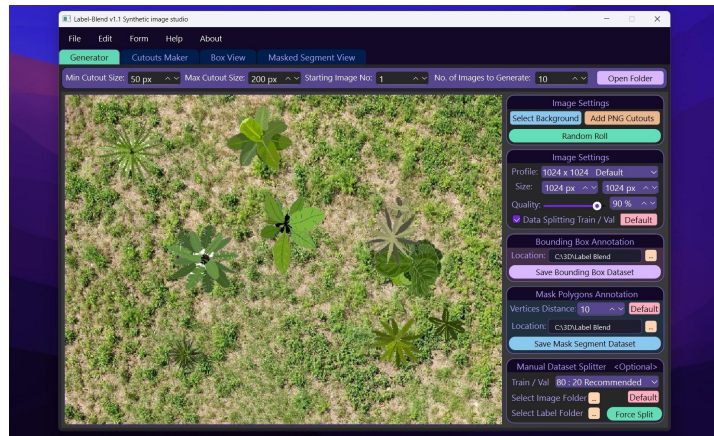


2. Biasing problem in class based bounding box labelling

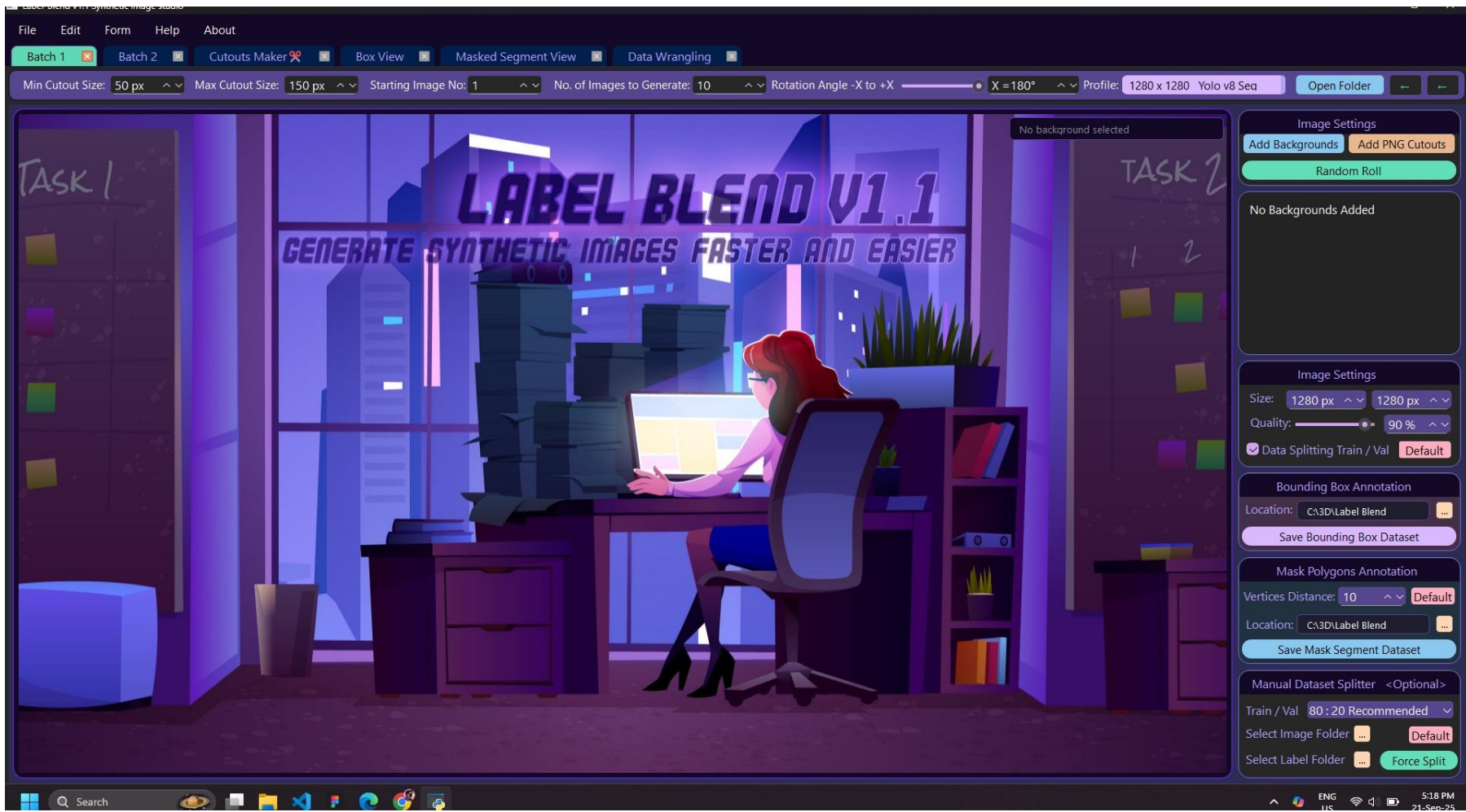


SOLUTION

Project image showing

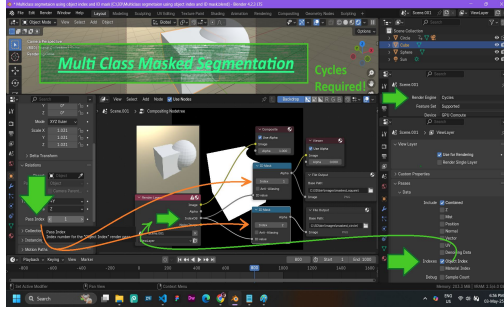


Label-Blend Studio, Main screen snapshot-

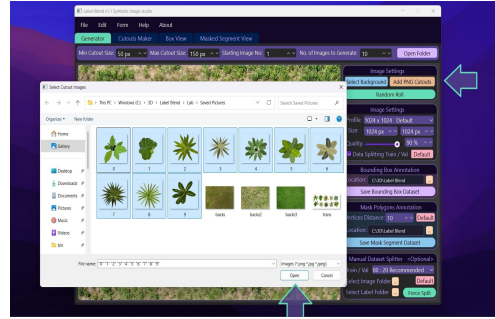


Process of labelling large image datasets using label-blend studio-

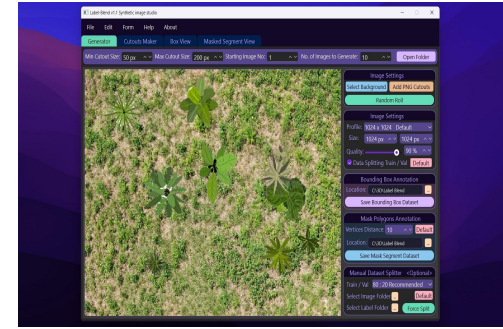
1. Creation of transparent cutouts



2. Loading cutouts into our project



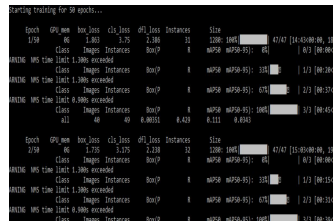
3. Loading background & Customising properties and generating the dataset



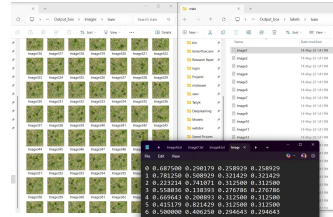
7. Deploy the model



6. Train the model



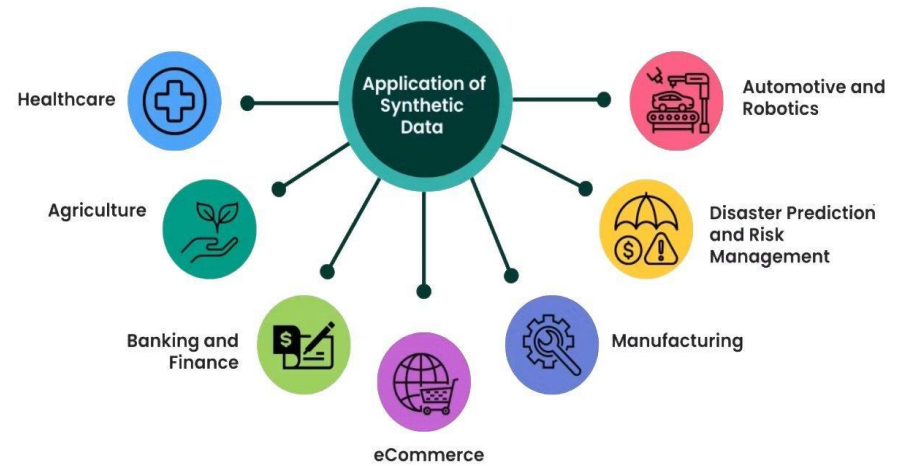
5. Labelled Data output



4. Check Annotations Accuracy



Future applications: & Infinite possibilities



1. Autonomous Vehicles: we can Train self-driving cars for object detection and road segmentation.

2. Medical Imaging: we can generate similar labeled medical scans for AI-driven diagnostics.

3. Augmented Reality (AR) & Virtual Reality (VR):
Security & Surveillance

4. Agriculture: we can generate synthetic data for ai model that Identify crop diseases using AI-powered image segmentation.

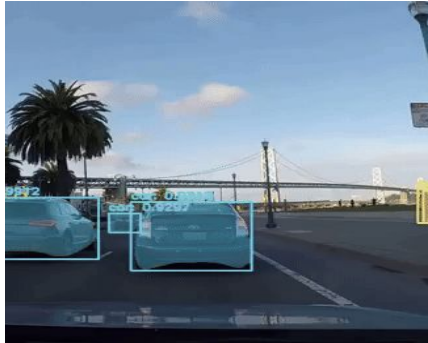
5. Robotics

6. Gaming & Animations

7. HealthCare

8. Manufacturing etc..

Autonomous Driving



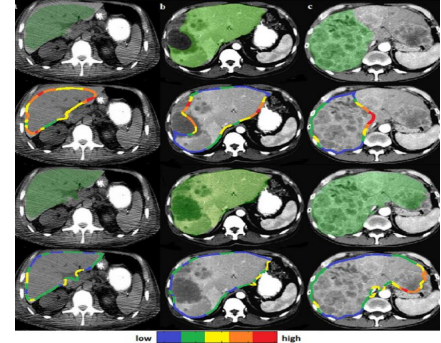
Robotics and vision



Virtual Reality



Healthcare and medicals



RESEARCH AND REFERENCES

- **A survey of synthetic data augmentation methods in computer vision**
 <https://arxiv.org/abs/2403.10075/>
- **Data augmentation: A comprehensive survey of modern methods**
 <https://www.sciencedirect.com/science/article/pii/S2590005622000911/>
- **What is YOLO v8: An In-Depth Exploration of ...**
 <https://arxiv.org/abs/2408.15857/>
- **Object Detection Using YOLO v8" (PDF)**
 <https://www.ijcrt.org/papers/IJCRT2412497.pdf/>
- **A Review of Synthetic Image Data and Its Use in Computer Vision**
 <https://pmc.ncbi.nlm.nih.gov/articles/PMC9698631/>

Prototype Video Link -  <https://youtu.be/TtHNoEp-X44>

Thank You

OIST, Bhopal